

Tutorial - Probability

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Proof of Bessel's Correction

Bessel's correction is the division of the sample variance by $N - 1$ rather than N . The proof shows why dividing by N is a biased estimate of population variance and how to correct it by dividing with $N - 1$, [Link](#).



Random Variables

Random Process: An Experiment which results in different outcomes each time it is repeated in a similar manner.

Random Variables: A function which assigns a numerical value to each possible outcome of a random experiment.

- ▶ **Discrete Random Variable:** A random variable that can take a countable number of possible values. Eg: let Y be the number of defective items found in a batch of 100.
- ▶ **Continuous Random Variable:** A random variable that can take any value within a continuous range. Eg: The temperature in Chennai at 2 PM — could be 31.2°C , 31.25°C , 31.251°C , etc.



Probability Mass Function (PMF)

Applies to: Discrete Random Variables

Definition: The PMF $P(X = x)$ gives the probability that a discrete random variable X takes the value x .

Properties:

- ▶ $P(X = x) \geq 0$ for all x
- ▶ $\sum_x P(X = x) = 1$

Example: Roll of a fair die: $P(X = x) = \frac{1}{6}$ for $x = 1, 2, 3, 4, 5, 6$



Probability Density Function (PDF)

Applies to: Continuous Random Variables

Definition: The PDF $f(x)$ describes how probability is distributed over values of X .

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

Properties:

- ▶ $f(x) \geq 0$ for all x
- ▶ $\int_{-\infty}^{\infty} f(x) dx = 1$
- ▶ Probability at a given point is zero

Example: Heights of students $\sim$ Normal distribution:

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$



Distributions

- ▶ Bernoulli
- ▶ Uniform
- ▶ Binomial
- ▶ Poisson
- ▶ Geometric
- ▶ Normal/ Gaussian
- ▶ Exponential
- ▶ Hypergeometric
- ▶ Negative Binomial

Note to Students: It is recommended to be familiar with the following for each distribution:

- ▶ PMF or PDF expression
- ▶ CDF (Cumulative Distribution Function)
- ▶ Mean and Variance formulas



Problem 1

You ask your neighbour to water a plant while you are on vacation. Without water it will die with probability of 0.8 and with water it will die with probability of 0.15. You are 90% sure that the neighbour will remember to water the plant.

- ▶ What is the probability that the plant will be alive when you return?
- ▶ If it is dead what is the probability that the neighbour forgot to water it ?



Problem 1 - Solution

Given:

- ▶ $P(W) = 0.9$ (Neighbour waters)
- ▶ $P(W') = 0.1$ (Neighbour forgets)
- ▶ $P(D | W) = 0.15$ (Plant dies if watered)
- ▶ $P(D | W') = 0.8$ (Plant dies if not watered)

Let: D = plant dies, $A = D' =$ plant alive

(a) Probability that the plant is alive:

$$P(A) = P(A | W)P(W) + P(A | W')P(W')$$

$$P(A | W) = 1 - 0.15 = 0.85, \quad P(A | W') = 1 - 0.8 = 0.2$$

$$P(A) = (0.85)(0.9) + (0.2)(0.1) = 0.785$$

(b) Probability neighbour forgot given plant is dead:

$$P(W' | D) = \frac{P(D | W')P(W')}{P(D)}$$

$$P(D) = (0.15)(0.9) + (0.8)(0.1) = 0.215$$

$$P(W' | D) = \frac{(0.8)(0.1)}{0.215} = 0.372$$

Final Results: $P(\text{Alive}) = 0.785$, $P(\text{Forgot} | \text{Dead}) = 0.372$



Problem 2

In a certain college, 4% of the boys and 1% of girls are taller than 1.8m. Further more 60% of the students are girls. If a student is selected at random and is found taller than 1.8m, what is the probability that the student is a girl ?



Problem 2 - Solution

Given:

$$\begin{aligned}P(\text{boy}) &= 0.4, & P(\text{girl}) &= 0.6, \\P(\text{tall} \mid \text{boy}) &= 0.04, & P(\text{tall} \mid \text{girl}) &= 0.01\end{aligned}$$

To find: $P(\text{girl} \mid \text{tall})$

Using Bayes' Theorem:

$$P(\text{girl} \mid \text{tall}) = \frac{P(\text{tall} \mid \text{girl}) P(\text{girl})}{P(\text{tall})}$$

Compute:

$$\begin{aligned}P(\text{tall}) &= P(\text{tall} \mid \text{girl})P(\text{girl}) + P(\text{tall} \mid \text{boy})P(\text{boy}) \\&= (0.01)(0.6) + (0.04)(0.4) \\&= 0.006 + 0.016 = 0.022\end{aligned}$$

Therefore:

$$P(\text{girl} \mid \text{tall}) = \frac{0.01 \times 0.6}{0.022} = \frac{0.006}{0.022} = 0.2727$$

$P(\text{girl} \mid \text{tall}) = 0.273 \text{ (approximately 27.3\%)}$



Problem 3

If 5 red and 8 blue balls are there in an urn and 6 red and 10 blue balls are there in the second

- ▶ If one ball is picked at random from one of these urns, what is the probability of it being red ?
- ▶ If the ball picked is blue what is the probability that it is picked from the first urn ?



Problem 3 - Solution

Given:

Urn 1: 5R, 8B (total 13)

Urn 2: 6R, 10B (total 16)

$$P(U_1) = P(U_2) = \frac{1}{2}$$

(a) Probability that the ball is red:

$$\begin{aligned} P(R) &= P(R|U_1)P(U_1) + P(R|U_2)P(U_2) \\ &= \frac{1}{2} \left(\frac{5}{13} \right) + \frac{1}{2} \left(\frac{6}{16} \right) = \frac{79}{208} \approx 0.380 \end{aligned}$$

(b) Probability that the ball came from Urn 1 given it is blue:

$$\begin{aligned} P(U_1|B) &= \frac{P(B|U_1)P(U_1)}{P(B)} \\ P(B) &= \frac{1}{2} \left(\frac{8}{13} \right) + \frac{1}{2} \left(\frac{10}{16} \right) = \frac{129}{208} \\ P(U_1|B) &= \frac{(8/13)(1/2)}{129/208} = \frac{832}{1677} \approx 0.496 \end{aligned}$$

$P(\text{Red}) = 0.380, \quad P(U_1 \text{Blue}) = 0.496$



Thanks

